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30 April 1984
SUPERSEDING
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MILITARY SPECIFICATION

TEST SET, POWER PANEL, TS-2661/TSM-100A

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of Power Panel Test Set hereinafter referred to as the "test set" which forms a part of the Electronics System Test Set type AN/TSM-100A, used on the Vulcan Air Defense Weapon System.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-5422	- Testing, Environmental, Aircraft Electronic Equipment
MIL-F-13926	- Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	- Parts, Equipment and Tools for Ordnance Material: Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QA Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-T-43435	- Tape: Impregnated, Lacing and Tying
MIL-C-46357	- Computer, Gun Direction; M18 Manufacturing Processes and Inspection of
MIL-C-60898	- Circuit Card Assemblies, General Specification for
MIL-C-60898/10	- Circuit Card Assemblies, Power Unit

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

F9333053	- Panel Assembly, Power
F9333068	- Schematic Diagram, Electrical, Power Panel, Test Unit

PACKAGING DATA SHEET

9333053	- Packaging of Test Set, Power Panel

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Fabrication. The test set shall be manufactured in accordance with Drawing F9333053 and drawings pertaining thereto.

3.2 General specifications. Where applicable, the test set shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishes

d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly. The requirements for etched circuit assemblies of MIL-C-46357 shall apply with modification as follows:

(1) Plated etched circuit boards. In addition, assemblies shall not contain copper or nickel visible through hot solder dip, when viewed at 10 times magnification.

(2) Location grid. The component mounting holes shall be within 0.007 inch of the true grid intersection, instead of 0.005.

(3) Transistor and diode leads. Transistor and metal cased diode leads shall be covered with sleeving in areas where there is a possibility of shorting. No color coding of sleeving shall be required.

(4) Drying. The drying process shall not be required.

b. Wiring. The requirements for wiring of MIL-C-46357 shall apply with modification as follows:

(1) Spot ties. Tape, Type 1, Finish B per MIL-T-43435 may be utilized without the need for varnish on the knots.

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(2) Wire termination. Removal of insulation by controlled heat methods shall be acceptable.

(3) Chassis cleaning. Compressed air or vacuum may be utilized for cleaning. Vibration shall not be used to clean the chassis.

c. Threaded fasteners. The requirements for threaded fasters of MIL-C-46357 shall apply except that maximum projection of screws shall be that of the next standard length screw beyond the minimum projection, unless electrical or mechanical interference exists.

d. Marking. The requirements for marking of MIL-C-46357 shall apply except that solvo photo etch process is acceptable.

3.2.3 Circuit card assemblies. The Circuit Card Assembly, Power Unit, shall meet the requirements of MIL-C-60898/10.

3.3 Environmental service conditions.

3.3.1 Vibration (non-operating). The test set shall be subjected to a simple harmonic motion having excursion as follows:

Frequency (Hz + 2 percent)	Excursion (double amplitude) (inches + 10 percent)
5 to 15	0.06
15 to 25	0.04
25 to 55	0.02

The frequency shall be varied logarithmically between 5 and 55 Hz within 15 minutes and returned to 5 Hz in 15 minutes. The motion shall be applied to the test set for a period of 30 minutes in each direction of the 3 mutually perpendicular axes (total of 90 minutes), relative to its normal operational position. Subsequent to vibration, the test set shall not be damaged or adversely affected and shall meet the electrical requirements of 3.4.

3.3.2 Temperatures

3.3.2.1 Operating. The test set shall meet the requirements of 3.4 while exposed to temperatures of +32°F and +131°F.

3.3.2.2 Storage. The test set shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +185°F.

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3.3.3 Humidity. The test set shall function as specified in 3.4 and shall not suffer any damage or deterioration which shall impair satisfactory performance after being subjected to ten consecutive humidity cycles at relative humidities between 85 and 95 percent and temperatures from 82° to 160°F over a period not less than 240 hours.

3.3.4 Shock. The test set shall show no evidence of damage or be adversely affected after being subjected to a total of 18 shocks consisting of three shocks in opposing directions along each of the three mutually perpendicular axes. Each shock shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11±1 millisecond.

3.3.5 Altitude

3.3.5.1 Operating. The test set shall not be damaged or adversely affected when exposed for a period of not less than four hours to conditions encountered at an altitude of 10,000 feet above sea level. During and subsequent to such exposure the test set shall meet the requirements of 3.4.

3.3.5.2 Non-operating. The test set shall meet the requirements of 3.4 after exposure to conditions encountered at 50,000 feet above sea level for a period of not less than one hour. The test set shall not be damaged or be adversely affected due to such exposure.

3.3.6 Sand and dust. The equipment shall withstand, in both an operating and non-operating condition, exposure to sand and dust particles as encountered in desert areas.

3.3.7 Salt spray. Equipment shall withstand, in both an operating and non-operating condition, exposure to salt-sea atmosphere.

3.3.8 Fungus-inert materials. The requirements for fungus-inert materials shall be in accordance with MIL-STD-454.

3.4 Electrical.

3.4.1 Continuity. The circuitry shall be in accordance with electrical schematic drawing 9333068. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein.

3.4.2 Functional

3.4.2.1 Conditions of operation. Unless otherwise specified, the requirements of 3.4.2 shall be met with the conditions shown in Figure 1. Also, unless otherwise specified, the DISTRIBUTION +24 Vdc, ON-OFF switch shall be in the "ON" position.

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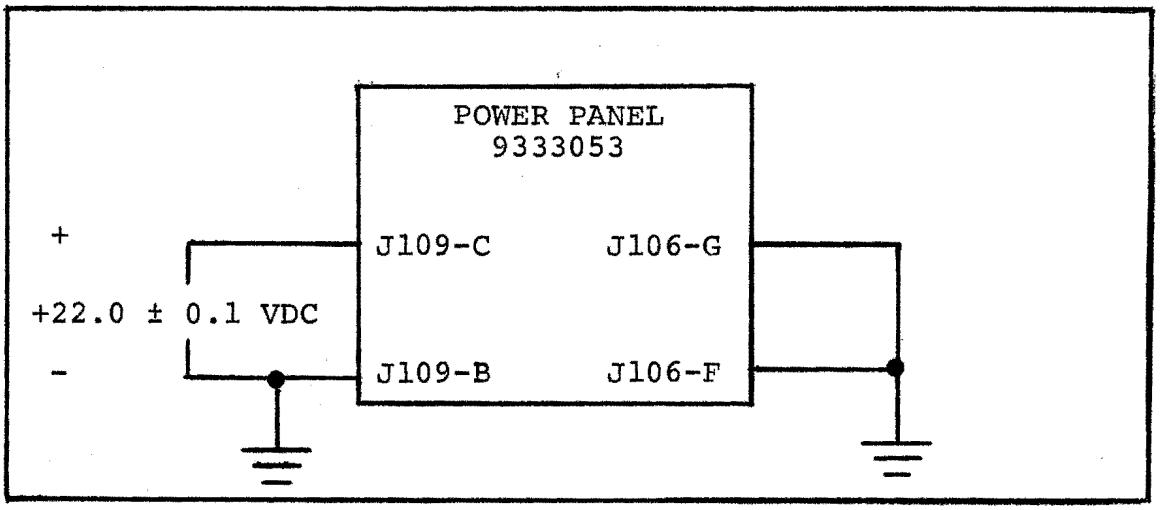


Figure 1. Operational test setup.

3.4.2.2 Distribution.

3.4.2.2.1 +24 Vdc, ON-OFF. The +24 Vdc indicator lamp shall LIGHT only when the DISTRIBUTION, +24 Vdc, ON-OFF switch is in the "ON" position and CB2 is closed. A potential of +22.0 ± 0.5 Vdc shall be present between the +24 Vdc, + test point (TP4) and GROUND only when the +24 Vdc, ON-OFF switch is in the "ON" position.

3.4.2.2.2 SIGHT CUR GEN. With the DISTRIBUTION switch in the "SIGHT CUR GEN" position, the UNIT No. 1 and INVERTER indicator lamps shall LIGHT. A potential of +22.0 ± 0.5 Vdc shall be present between J105-f and GROUND (TP-11) only when in this switch position.

3.4.2.2.3 DB. With the DISTRIBUTION switch in the "DB" position, the UNIT NO. 1 indicator lamp shall LIGHT. A potential of +22.0 ± 0.5 Vdc shall be present between J105-Y and GROUND only when in this switch position.

3.4.2.2.4 SVO AMPL. A potential of +22.0 ± 0.5 shall be present between J105-t and GROUND only when the DISTRIBUTION switch is in the "SVO AMPL" position.

3.4.2.2.5 CONT PNL. A potential of +22.0 ± 0.5 Vdc shall be present between J105-m and GROUND only when the DISTRIBUTION switch is in the "CONT PNL" position.

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3.4.2.2.6 AN/MWM-2. The resistance between J107-U and J107-E shall be less than 2.5 ohms only when the DISTRIBUTION switch is in the "AN/MWM-2" position. In all other DISTRIBUTION switch positions the resistance shall be greater than 500 K ohms.

3.4.2.2.7 Unit No. 1. The UNIT NO. 1 indicator lamp shall LIGHT when the DISTRIBUTION switch is in the "UNIT NO. 1" position.

3.4.2.2.8 115V, 400CY, 1 PHASE. The 115V, 400CY, 1 PHASE indicator lamp shall LIGHT only when 115 volts RMS ±10 percent is applied between J108-E and J108-D and the 115V, 400CY, 1 PHASE ON-OFF switch (S4) is in the "ON" position and CB4 is closed. Measure 115V, 400CY, 1 Phase across J10 pins A and B.

3.4.2.2.9 RADAR NOISE. The UNIT NO. 1 INVERTER indicator lamp shall LIGHT when in the RADAR NOISE position. The voltage between J105-F and J106-O and J106-R shall be +24 Vdc.

3.4.2.3 K3 relay. With +22.0 ± 0.1 Vdc applied between J106-C and GROUND and the DISTRIBUTION switch in the "+24 Vdc ONLY" position, the INVERTER indicator lamp shall LIGHT.

3.4.2.4 K4 relay. The resistance between the pins specified in Table I shall be as specified in Table I when no voltage is applied to J106-B. When +22.0 ± 0.1 Vdc is applied between J106-B and GROUND, the resistance between the pins shall be specified in Table II.

TABLE I. Resistance values with no voltage.

Pin to Pin	Resistance
J107-W to J106-I	greater than 500 K ohms
J107-R to J106-J	greater than 500 K ohms
J107-U to J107-E	greater than 500 K ohms
J107-U to J105-a	less than 2.5 ohms
J107-V to J106-$\overline{Z}$	greater than 500 K ohms
J107-V to J105-S	less than 2.5 ohms
J107-X to J106-a	greater than 500 K ohms
J107-X to J105-$\overline{T}$	less than 2.5 ohms

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TABLE II. Resistance values with 22 Vdc applied.

Pin to Pin	Resistance
J107-W to J106-I	less than 2.5 ohms
J107-R to J106-J	less than 2.5 ohms
J107-U to J107-E	less than 2.5 ohms
J107-U to J105-a	greater than 500 K ohms
J107-V to J106-$\overline{Z}$	less than 2.5 ohms
J107-V to J105-S	greater than 500 K ohms
J107-X to J106-a	less than 2.5 ohms
J107-X to J105-$\overline{T}$	greater than 500 K ohms

3.4.2.5 K5 relay. The resistance between J107-K and J105-b shell be less than 2.5 ohms, and the resistance between J107-K and shall be greater than 500 K ohms when no voltage is applied to When +22.0 ±.1 Vdc is applied between J106-A and GROUND, the resistance between J107-K and J105-b shall be greater than 500 K ohms, and the resistance between J107-K and J106-L shall be less than 2.5 ohms.

3.4.2.6 -OUTPUT ADJUST. With -15.0 ±.1 Vdc applied between J107-R(-) and J107-C(+), the voltage between the two -OUTPUT TO AN/MWM-2 test points (TP12 and TP13) and GROUND (TP10) shall be 0.0 ± 0.1 Vdc when the -OUTPUT ADJUST knob is in the "0.00" position. The voltage shall be -15.0 ± 0.1 Vdc when the knob is in the "10.00" position.

3.4.2.7 +OUTPUT ADJUST. With +15.0 ± 0.1 Vdc applied between J107-W (+) and J107-C(-), the voltage between the two +OUTPUT TO AN/MWM-2 test points (TP14 and TP15) and GROUND (TP10) shall be 0.0 ± 0.1 Vdc when the +OUTPUT ADJUST knob is in the "0.00" position. The voltage shall be +15.0 ± 0.1 Vdc when the knob is in the "10.00" position.

3.4.2.8 AN/MWM-2 VARIABLE FREQUENCY SQUARE WAVE. With the DISTRIBUTION switch in the "AN/MWM-2" position and with a 10 K ohm, ± 5 percent ½ watt resistor between the VARIABLE FREQUENCY SQUARE WAVE, + (TP1) and GRD (TP2) test points, the requirements of 3.4.2.8.1 and 3.4.2.8.2 shall be met.

3.4.2.8.1 High frequency. The voltage between the + and GRD test points shall be as shown in Figure 2 when the FREQUENCY knob is adjusted clockwise and the pulse width shall be adjusted by using the PULSE WIDTH knob to parameters of Figure 2.

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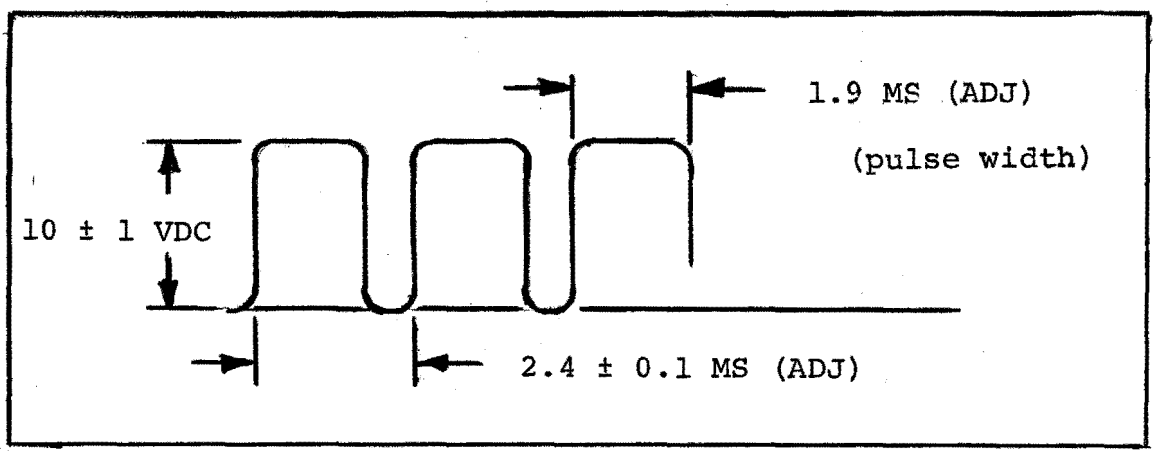


Figure 2. High frequency voltage output.

3.4.2.8.2 Low frequency. With +28.0 ± 0.1 Vdc applied between J109-C (+) and J109-B(-) instead of +22.0 ± 0.1 Vdc, the voltage between + and GRD test points shall be as shown in Figure 3 when the FREQUENCY knob is in the fully counterclockwise position. The pulse width shall be obtained by adjusting the PULSE WIDTH knob.

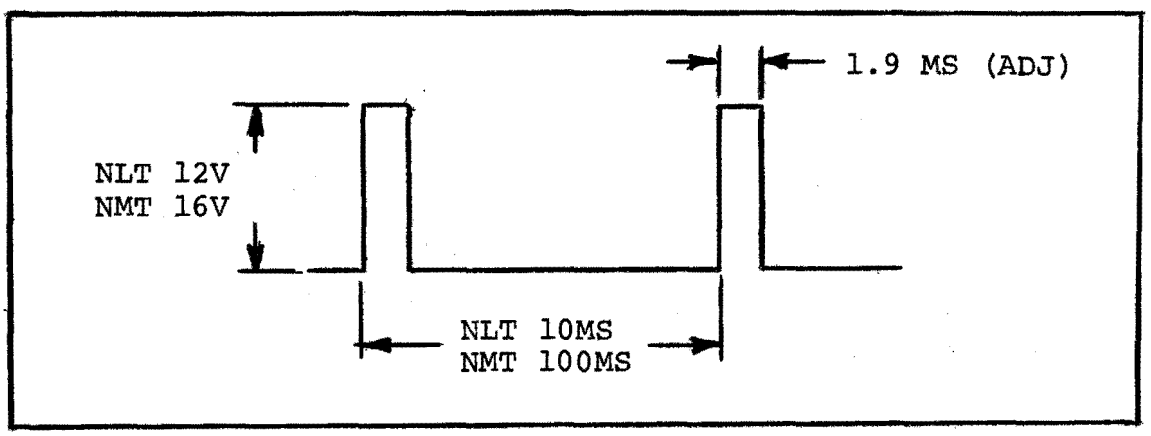


Figure 3. Low frequency voltage output.

3.4.2.9 Resistance. The resistance between J107-M and J107-N and also between J107-M and J107-P shall be 267 ohms ± 3 percent.

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3.4.2.10 CR1 diode.

3.4.2.10.1 Reverse current. With +28.0 ± 0.5 Vdc applied between J109-C(+) and J108-F (-), the current shall be less than 1 milliampere through these two pins.

3.4.2.10.2 Forward drop. With 20.0 ± 0.1 amperes applied between J109-C(-) and J108-F(+), the voltage between these two pins shall be less than +1.50 Vdc.

3.4.2.11 10 Amperes circuit breaker. The circuit breaker shall open when subjected to a current greater than 10 amperes. The maximum test overload shall never be greater than 200 percent of the circuit breaker's rated value. At the maximum overload condition, the circuit breaker shall open within 60 seconds.

3.4.2.12 1 Ampere circuit breaker. The circuit breaker shall open when subjected to a current greater than 1 ampere. The maximum test overload shall never be greater than 200 percent of the circuit breaker's rated value. At the maximum overload condition, the circuit breaker shall open within 60 seconds.

3.4.2.13 5 Amperes circuit breaker. The circuit breaker shall open when subjected to a current greater than 5 amperes. The maximum test overload shall never be greater than 200 percent of the circuit breaker's rated value. At the maximum overload condition, the circuit breaker shall open within 60 seconds.

3.4.2.14 50 Amperes circuit breaker. The circuit breaker shall open when subjected to a current greater than 50 amperes. The maximum test overload shall never be greater than 200 percent of the circuit breaker's rated value. At the maximum overload condition, the circuit breaker shall open within 60 seconds.

3.4.3 Electrical performance at temperature extremes. The following requirements shall apply for the conditions of 3.3.2 (as applicable).

3.4.3.1 Square Wave, high frequency. For the conditions of 3.4.2.8.1 the output shall be the same as shown in Figure 2 except the duty cycle shall not exceed 2.5 milliseconds and the pulse height shall be +10 ± 2 Vdc.

3.4.3.2 Square Wave, low frequency. For the conditions of 3.4.2.8.2, the output shall be the same as shown in Figure 3 except the duty cycle shall not be less than 10 milliseconds and the pulse height shall be +14 ± 3 Vdc.

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3.4.3.3 CR1 diode. For the conditions of 3.4.2.10, the reverse current shall be less than 1.5 milliamperes and the forward drop shall be less than +2.0 Vdc.

3.4.4 Wire (hookup). Standard conductors of suitable degrees of flexibility shall be used whenever practicable within the requirements of this specification. Hookup wire shall be of such cross section and temper as to provide ample and safe current-carrying capacity and mechanical strength. In general, wire shall not be smaller than AWG No. 22. Wire directly associated with the rimary power circuit shall be of a size consistent with the current requirements of the circuit. When it can be demonstrated that the use of smaller hookup wire will result in no loss of performance and at the same time will produce benefits, such as reduction in weight or improved accessibility, smaller wire may be used. This provision is made specifically to permit the use of smaller wire in cables having a large number of wires and having adequate support against adverse effects of vibration. The size of wire leads supplied with parts shall be controlled by the specifications for these parts.

3.4.4.1 Insulation on conductors. Insulation of the sleeve type may be used, provided its length is sufficient to prevent slippage along the conductor to such an extent as to expose a section of the conductor that should be insulated. Insulation shall provide adequate dielectric strength and adequate leakage resistance when the equipment is operated under the designated service conditions. The mechanical and electrical properties of the insulating materials and impregnating compounds shall not be permanently impaired as a result of being exposed to extremes of temperature incurred in the equipment. Rubber without braid may be used only when the insulation is not depended upon to space or support the conductor, or when a number of conductors are assembled in the form of cordage and the assembly provided with a braid covering which has been suitably varnished or lacquered. All conductor insulation shall be of the noncombustible or slow-burning type; that is, with the insulated conductor held in a horizontal position in still air, self-sustained combustion of the insulation or lacquer shall not progress at a rate in excess of 1 inch in 1 minute.

3.5 First Article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.6 Workmanship. The requirements of MIL-STD-454 and other applicable specifications shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for

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the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Unless otherwise specified in the detail specification, or contract, the following types of inspection shall be conducted on the unit or product:

a. First article inspection (4.3)

b. Quality conformance inspection (4.4)

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer (see 6.2) for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 3 test sets.

4.3.2 Inspection to be performed. First article samples may be subjected by the Government to any or all of the examinations and tests specified in 4.4.2 and 4.4.4 of this specification and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless

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of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests. Except where otherwise specified herein, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables III and IV and methods of inspecion in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.4.2.1 Components and subassemblies. Printed circuit subassemblies shall be inspected in accordance with the inspection provisions of MIL-C-60898. All other components and subassemblies shall be inspected to the applicable requirements of MIL-F-13926.

4.4.2.2 Control tests. One test set shall be selected at random as a control sample from each 100 produced. The control samples shall meet the requirements and tests of Table III after they have met the requirements and tests of Table IV.

TABLE III. Control tests

Test	Requirement	Test Procedures
Fabrication	3.1	Applicable drawings - visual
General Specifications	3.2	Visual-tactile
Vibration	3.3.1	4.5.1.1, 4.5.1.2
Temperature	3.3.2, 3.4.3	4.5.1.1, 4.5.1.4
Humidity	3.3.3	4.5.1.1, 4.5.1.3
Shock	3.3.4	4.5.1.1, 4.5.1.5
Altitude	3.3.5	4.5.1.1, 4.5.1.6
Sand and dust	3.3.6	4.5.1.1, 4.5.1.7
Salt spray	3.3.7	4.5.1.1, 4.5.1.8
Fungus	3.3.8	4.5.1.1, 4.5.1.9

4.4.3 Rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4.4 Functional tests. The requirements and tests in Table IV shall be inspected on a 100% basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature.	+50° to +90°F
b.	Relative humidity.	90% RH max.
c.	Pressure.	28 to 32 inches of Hg

TABLE IV. Functional tests.
(100 percent inspection)

	Characteristic	Requirement	Procedure
101	Continuity test	3.4.1	4.5.2.1
102	Conditions of operation	3.4.2.1	4.5.2.2.1
103	+24 Vdc, ON-OFF	3.4.2.2.1	4.5.2.2.2.1
104	SIGHT CUR GEN	3.4.2.2.2	4.5.2.2.2.2
105	DB	3.4.2.2.3	4.5.2.2.2.3
106	SVO AMPL	3.4.2.2.4	4.5.2.2.2.4
107	CONT PNL	3.4.2.2.5	4.5.2.2.2.5
108	AN/MWM-2	3.4.2.2.6	4.5.2.2.2.6
109	UNIT NO. 1	3.4.2.2.7	4.5.2.2.2.7
110	115, 400CY, 1 PHASE	3.4.2.2.8	4.5.2.2.2.8
111	RADAR NOISE	3.4.2.2.9	4.5.2.2.2.9
112	K3 relay	3.4.2.3	4.5.2.2.3
113	K4 relay	3.4.2.4	4.5.2.2.4
114	K5 relay	3.4.2.5	4.5.2.2.5
115	-OUTPUT ADJUST	3.4.2.6	4.5.2.2.6
116	+OUTPUT ADJUST	3.4.2.7	4.5.2.2.7
117	AN/MWM-2 VARIABLE FREQUENCY SQUARE WAVE	3.4.2.8	4.5.2.2.8
118	High frequency	3.4.2.8.1	4.5.2.2.8.1
119	Low frequency	3.4.2.8.2	4.5.2.2.8.2
120	Resistance	3.4.2.9	4.5.2.2.9
121	Reverse current	3.4.2.10.1	4.5.2.2.10.1
122	Forward drop	3.4.2.10.2	4.5.2.2.10.2
123	Circuit breaker, 10 amperes	3.4.2.11	4.5.2.2.11
124	Circuit breaker, 1 ampere	3.4.2.12	4.5.2.2.12
125	Circuit breaker, 5 amperes	3.4.2.13	4.5.2.2.13
126	Circuit breaker, 50 amperes	3.4.2.14	4.5.2.2.14
127	Electrical performance at temperature extremes	3.4.3	4.5.2.3
128	Square wave, high frequency	3.4.3.1	4.5.2.3.1
129	Square wave, low frequency	3.4.3.2	4.5.2.3.2
130	CR1 diode	3.4.3.3	4.5.2.3.3

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4.4.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the referenced specifications and standards cited in 4.4.2 and 4.4.4. Inspection equipment shall be as listed in Table V. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See 6.3 of this specification.

TABLE V. Inspection equipment list.

Item		Description
1.	POWER SUPPLY	0-30 Vdc, adjustable. Load regulation of 1% or better from 0 to 5 amperes. Less than 1% ripple.
2.	DIGITAL VOLTMETER	Accuracy and resolution of ± 0.2% from 1 to 100 Vdc and ± 0.2% from 1 to 200 Vac, 400 Hz. Input impedance 10 megohms or greater.
3.	OHMMETER	Accuracy and resolution of ± 0.2% from 1 ohm to 500 K ohms.
4.	POWER SUPPLY	115 Vac ± 10%, 400 cycles, 1 phase, 500 milliamperes.
5.	POWER SUPPLY	0-30 Vdc adjustable. 0-20 amperes adjustable. Less than 1% ripple.
6.	RESISTOR	10 K ohms, 5%, ½ watt
7.	OSCILLOSCOPE	Capable of observing and measuring 0 to 360V peak to peak signals to within 1%, with a 3% time accuracy. Rise time: 100 nanoseconds or less. Automatic sweep speeds from 1 microsecond to 100 milliseconds.
8.	AMMETER	Accuracy and resolution of 2% or better from 0.02 milliampere to 20 ampere dc.
9.	POWER SUPPLY	A dc power supply and an ammeter indicating load current and an adjustable current limiting control. The capacity will be 100 amperes or greater.

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4.5 Methods of inspection.

4.5.1 Environmental

4.5.1.1 Test conditions. The test set shall be mounted in an upright position in a manner similar to that in which it will be in service and shall be subjected to the environmental tests of the following subparagraphs. The atmospheric pressure shall be room ambient. Except for the high and low temperature tests or the humidity tests, as applicable, the temperature and humidity shall be room ambient.

4.5.1.2 Vibration. The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1. Subsequent to the vibration test, the test set shall show no damage or defect resulting from the test when examined by visual and tactile means. The test set shall then be tested to determine compliance with requirements specified under 3.3.1.

4.5.1.3 Humidity. The humidity test shall be performed in accordance with Procedure I of Method 507 in MIL-STD-810. Subsequent to the humidity test, the test set shall be tested to determine compliance with 3.3.3.

4.5.1.4 Temperature. Test in accordance with Procedure I of Method 501 in MIL-STD-810 for high temperature operating and non-operating under conditions specified in 3.3.2. Test in accordance with Procedure I of Method 502 for low temperature operating and non-operating under conditions specified in 3.3.2, and verify requirements of 3.3.2.1 and 3.3.2.2.

4.5.1.5 Shock. The shock test shall be performed in accordance with MIL-T-5422.

4.5.1.6 Altitude. The altitude test shall be performed in accordance with the following steps:

Step 1. Place the test set in the test chamber in a manner that will simulate service usage, making connections and attaching onstrumentation as necessary. Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. For tests where temperature values

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are controlled, the test chamber shall be at standard ambient conditions when the test set is installed. The test set shall then be operated to determine that no malfunction or damage was caused due to faulty installation or handling. The requirement for operation following installation of the test set in the test facility is applicable only when operation is required during or immediately following exposure to the specified test. Standard ambient temperature shall be maintained during the entire test.

Step 2. Reduce the chamber internal pressure to 87.5 mm of Hg (3.44 in. of Hg or 50,000 ft. above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than one hour.

Step 3. Increase the chamber pressure to 523 mm of Hg (20.6 in. of Hg or 10,000 ft. above sea level) and then operate the test set and compare the results with the data obtained in accordance with pretest performance record made of all data necessary to determine compliance with required performance.

Step 4. With the test set not operating, return the chamber to standard ambient conditions.

Step 5. Operate the test set and compare the data obtained with the data obtained in accordance with pretest performance record.

Step 6. The test set shall be visually inspected and a record made of any damage or deterioration resulting from the test. If a test chamber is used for the test, perform a visual inspection of the test within the chamber at test conditions, when possible. Upon completion of the test, visually inspect the test set again after the test set has been returned to standard ambient conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external parts which in any manner prevent the test set from meeting operational service or maintenance requirements shall provide reason to consider the test set as having failed to withstand the conditions of the test.

4.5.1.7 Sand and dust. The sand and dust test shall be performed in accordance with MIL-T-5422.

4.5.1.8 Salt spray. The salt spray test shall be performed in accordance with MIL-T-5422.

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4.5.1.9 Fungus. The fungus resistance test shall be performed in accordance with MIL-T-5422.

4.5.2 Electrical tests

4.5.2.1 Continuity test. A continuity test shall be performed to verify conformance to 3.4.1.

4.5.2.2 Functional tests

4.5.2.2.1 Conditions of operation. Connect the POWER SUPPLY, item 1 of 4.4.5, between J109-C (+) and J109-B and set for an output voltage of + 22.0 ± 0.1 Vdc. Connect J106-G and J106-F to J109-B. Set the DISTRIBUTION, +24 Vdc, ON-OFF switch in the "ON" position. Unless otherwise specified, all subsequent tests will be performed with these conditions of operation.

4.5.2.2.2 DISTRIBUTION

4.5.2.2.2.1 +24 Vdc, ON-OFF switch. Connect the DIGITAL VOLTMETER, Item 2 of 4.4.5, between +24 Vdc, + test point (TP4) and GRD. Operate the DISTRIBUTION, +24 Vdc, ON-OFF switch and the test point +24 Vdc, ON-OFF switch and verify conformance with 3.4.2.2.1 when CB2 is closed.

4.5.2.2.2.2 SIGHT CUR GEN. Connect the DIGITAL VOLTMETER, item 2 of 4.4.5, between J105-f and GROUND. Rotate the DISTRIBUTION switch to all positions and verify conformance with 3.4.2.2.2.

4.5.2.2.2.3 DB. Connect the DIGITAL VOLTMETER, item 2 of 4.4.5, between J105-Y and GROUND. Rotate the DISTRIBUTION switch to all positions and verify conformance with 3.4.2.2.3.

4.5.2.2.2.4 SVO AMPL. Connect the DIGITAL VOLTMETER, item 2 of 4.4.5 between J105-t and GROUND. Rotate the DISTRIBUTION switch to all positions and verify conformance with 3.4.2.2.4

4.5.2.2.2.5 CONT PNL. Connect the DIGITAL Voltmeter, item 2 of 4.4.5, between J105-m and GROUND. Rotate the DISTRIBUTION switch to all positions and verify conformance with 3.4.2.2.5.

4.5.2.2.2.6 AN/MWM-2. Connect the OHMETER, item 3 of 4.4.5, between J107-U and J107-E. Rotate the DISTRIBUTION switch to all positions and set the +24 Vdc, ON-OFF switch (S3) in the "ON" position when the DISTRIBUTION switch is in the "AN/MWM-2" position. Verify that the resistance readings conform to the requirements of 3.4.2.2.6.

4.5.2.2.2.7 UNIT NO. 1. Set the DISTRIBUTION switch in the "UNIT NO. 1" position and verify that the UNIT NO. 1 indicator lamp LIGHTS.

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4.5.2.2.2.8 115V, 400CY, 1 PHASE. Connect the POWER SUPPLY, item 4 of 4.4.5, between J108-E and J108-D. Operate the 115V, 400CY, 1 PHASE, ON-OFF switch (S4) to verify conformance with 3.4.2.2.8 when CB4 is closed. Connect DIGITAL VOLTMETER, item 2 of 4.4.5, between J10-A and J10-B to verify conformance with 3.4.2.2.8.

4.5.2.2.2.9 RADAR NOISE. Set the DISTRIBUTION switch in the RADAR NOISE position and verify that UNIT NO. 1 and indicator lamp LIGHTS. Connect the Digital Voltmeter, item 2 of 4.4.5, between J105-F, positive lead and J106-0 and J106-R, negative leads and verify conformance with 3.4.2.2.9.

4.5.2.2.3 K3 relay. Using the POWER SUPPLY, item 5 of 4.4.5, apply +22.0 ± 0.1 Vdc between J106-C and GROUND. Set the DISTRIBUTION switch in the "+24 VDC ONLY" position and verify that the INVERTER indicator lamp LIGHTS.

4.5.2.2.4 K4 relay. Using the OHMMETER, item 3 of 4.4.5, measure the resistance of each "pin to pin" listing in Table I. Verify conformance with 3.4.2.4 with no voltage applied between J106-B and GROUND. Using the POWER SUPPLY, item 5 of 4.4.5, apply +22.0 ± 0.1 Vdc between J106-B (+) and GROUND. Using the OHMMETER, measure the resistance of each "pin to pin" listing in Table II to verify conformance with 3.4.2.4.

4.5.2.2.5 K5 relay. Using the OHMMETER, item 3 and POWER SUPPLY, item 5 of 4.4.5, measure the resistance between J107-K and J106-L, and between J107-K and J105-b. Verify that the resistance conforms to the requirements of 3.4.2.5 with +22.0 ± 0.1 Vdc applied between J106-A and GROUND and also with the voltage removed.

4.5.2.2.6 -OUTPUT ADJUST. Using the POWER SUPPLY, item 5 of 4.4.5, apply +15.0 ± 0.1 Vdc between J107-C (+) and J107-R. Using the DIGITAL VOLTMETER, item 2 of 4.4.5, verify that the output voltage between the two -OUTPUT TO AN/MWM-2 test points and GROUND conform to 3.4.2.6.

4.5.2.2.7 +OUTPUT ADJUST. Using the POWER SUPPLY, item 5 of 4.4.5, apply +15.0 ± 0.1 Vdc between J107-W (+) and J107-C. Using the DIGITAL VOLTMETER, item 2 of 4.4.5, verify that the output voltage between the two, +OUTPUT TO AN/MWM-2, test points and GROUND (TP10) conform to 3.4.2.7.

4.5.2.2.8 AN/MWM-2 VARIABLE FREQUENCY SQUARE WAVE. Connect the RESISTOR, item 6 of 4.4.5, between the AN/MWM-2 VARIABLE FREQUENCY SQUARE WAVE, + and GRD test points. Connect the OSCILLOSCOPE, item 7 of 4.4.5, vertical input across the RESISTOR. With these conditions, perform the tests of 4.5.2.2.8.1 and 4.5.2.2.8.2.

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4.5.2.2.8.1 High Frequency. Adjust the FREQUENCY knob and the PULSE WIDTH knob and observe the output pulse on the OSCILLOSCOPE to verify the requirements of 3.4.2.8.1.

4.5.2.2.8.2 Low Frequency. Increase the POWER SUPPLY, item 1 of 4.4.5, voltage to +28.0 ± 0.1 Vdc and set the FREQUENCY knob in the fully clockwise position. Observe the output pulse on the OSCILLOSCOPE. Vary the PULSE WIDTH knob and verify the requirements of 3.4.2.8.2.

4.5.2.2.9 Resistance. Using the OHMMETER, item 3 of 4.4.5, measure the resistance between J107-M and J107-P, and between J107-M and J107-N. Verify conformance to 3.4.2.9.

4.5.2.2.10 CR1 diode

4.5.2.2.10.1 Reverse current. Connect the POWER SUPPLY, item 5, and the AMMETER, item 8 of 4.4.5, as shown in Figure 4. Set the power supply voltage to +28.0 ± 0.5 Vdc and apply the output to J109-C (+) and J108-F (-). Verify that the ammeter reading is less than 1 milliampere.

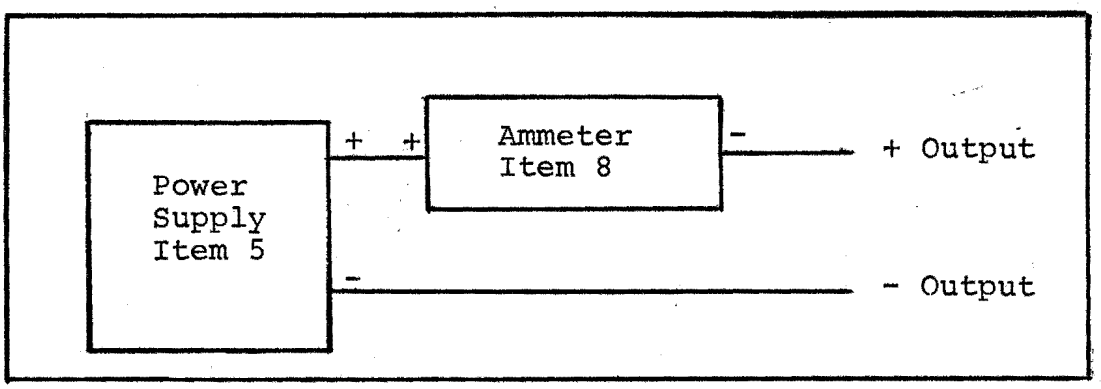


Figure 4. Test setup.

4.5.2.2.10.2 Forward drop. Connect the POWER SUPPLY, item 5, and the AMMETER, item 8 of 4.4.5, to J109-C (-) and J108-F (+). Adjust the power supply output to obtain an ammeter reading of 20.0 ± 0.1 amperes. Using the DIGITAL VOLTMETER, item 2 of 4.4.5, verify that the D.C. voltage between J109-C (-) and J108-F (+) is less than +1.50 Vdc.

4.5.2.2.11 10 Amperes. Connect the output of the POWER SUPPLY, item 9 of 4.4.5, across the terminals of the breaker under test. Turn on the POWER SUPPLY and increase its current output until the breaker under test opens. Verify that the breaker under test trips open with a current output greater than 10 amperes.

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4.5.2.2.12 1 Ampere. Connect the output of the POWER SUPPLY, item 9 of 4.4.5, across the terminals of the breaker under test. Turn on the POWER SUPPLY and increase its current output until the breaker under test opens. Verify that the breaker under test trips open with a current output greater than 1 ampere.

4.5.2.2.13 5 Amperes. Connect the output of the POWER SUPPLY, item 9 of 4.4.5, across the terminals of the breaker under test. Turn on the POWER SUPPLY and increase its current output until the breaker under test opens. Verify that the breaker under test trips open with a current output greater than 5 amperes.

4.5.2.2.14 50 Amperes. Connect the output of the POWER SUPPLY, item 9 of 4.4.5, across the terminals of the breaker under test. Turn on the POWER SUPPLY and increase its current output until the breaker under test opens. Verify that the breaker under test trips open with a current output greater than 50 amperes.

NOTE: The tests of 4.5.2.2.11, 4.5.2.2.12, 4.5.2.2.13 and 4.5.2.2.14 may be performed prior to the installation of the circuit breakers into the next higher assembly when verification of the tests are provided.

4.5.2.3 Electrical performance at temperature extremes. The following tests will be performed at the test conditions of 4.5.1.1 and 4.5.1.4 as applicable to determine conformance with 3.4.3.

4.5.2.3.1 Square wave, high frequency. Perform the procedures of 4.5.2.2.8.1 except verify that the output waveform conforms to the requirements of 3.4.3.1.

4.5.2.3.2 Square wave, low frequency. Perform the procedures of 4.5.2.2.8.2 except verify that the output waveform conforms to the requirements of 3.4.3.2.

4.5.2.3.3 CR1 diode. Perform the procedures of 4.5.2.2.10.1 and 4.5.2.2.10.2 except verify the reverse current and forward drop conform to the requirements of 3.4.3.3.

4.5.3 Wire insulation. Wire insulation requirements per 3.4.4 shall be verified by certification.

5. PACKAGING

5.1 General. Preservation, packaging, packing, and marking shall be in accordance with Packaging Data Sheet 9333053.

6. NOTES

6.1 Intended use. This test set is designed for rackmounting

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into the AN/TSM-100 Electronics System Test Set, which is used for troubleshooting and performance checking of specific weapon system components.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this and the item detail specifications.

b. Applicable stock number.

c. A block of serial numbers to identify each test set.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First Article Sample requirements, if other than specified in Section 4 of item detail specification.

f. Packaging requirements, if other than specified in Section 5 of item detail specification.

g. Certificate of conformance for each lot or shipment of product.

6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs conforming to Data Item Description DI-R-1714, tailored, with appropriate addendum to each. See 6.3.

b. Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410.

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government designs and contractor designs. Government designs are designated by drawing numbers in the appropriate paragraph in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government design is provided; however, unless otherwise specified, the contractor may submit alternate or modified contractor designs in accordance with 6.3.2 and 6.3.3 should he elect to do so.

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6.3.1 Government designs. Government designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in DOD-STD-100.

c. An envelope drawing, as defined in DOD-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which Government designs are not specified and include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with Government designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The item detail specification number, paragraph number and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAF-I (D), Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as US Army Armament Research and Development Command,

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(ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A199)

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